

A demonstration of the `achemso` $\LaTeX$ class[†]

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Abstract

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K-REACH; genotoxicity; in silico; QSAR; structural alerts; machine learning

Acknowledgement

Please use “The authors thank ...” rather than “The authors would like to thank ...”.

The author thanks Mats Dahlgren for version one of `achemso`, and Donald Arseneau for the code taken from `cite` to move citations after punctuation. Many users have provided feedback on the class, which is reflected in all of the different demonstrations shown in this document.

[†]A footnote for the title

Introduction

Genotoxicity is a critical toxicological endpoint that assesses the potential of chemicals to induce DNA damage, mutations, or carcinogenesis and plays a central role in regulatory chemical evaluation.^{1 2 3} Within the OECD genetic toxicology framework, the bacterial reverse mutation assay (Ames test; OECD Test Guideline 471) continues to serve as a cornerstone for detecting point mutations, while mammalian *in vitro* and *in vivo* assays provide complementary evidence for broader forms of genetic damage. In parallel, ICH S2(R1) recommends that bacterial, mammalian *in vitro*, and follow-up *in vivo* studies should be interpreted within an integrated testing strategy rather than treated as interchangeable endpoints. Accordingly, genotoxicity prediction is not a single-endpoint problem, and differences in assay biology, protocol design, and regulatory interpretation must be considered when constructing computational datasets and models.^{4 5 6}

The rapid expansion of publicly accessible toxicological data and the widespread adoption of machine learning have stimulated intense interest in *in silico* genotoxicity prediction.⁷

However, in this area, predictive performance is often constrained less by algorithmic sophistication than by the quality, provenance, and consistency of the underlying data. Classical cheminformatics studies have repeatedly shown that unresolved salts and mixtures, inconsistent parent-child representations, duplicate structures, erroneous structure records, and conflicting annotations can materially distort descriptor calculation, training distributions, and external validation results. Subsequent work has extended this argument from chemical structure curation to integrated chemical-biological curation, emphasizing that reproducible modeling requires explicit workflows for detecting duplicate records, reconciling inconsistent labels, and standardizing structure representations before any model development begins.^{8 9} For assay-driven toxicology datasets, the problem is even more acute: endpoint definitions, activity thresholds, cytotoxicity filters, signal interference, and purity criteria can all alter what is ultimately learned by a statistical model.¹⁰

These issues are particularly relevant in genotoxicity, where historical databases fre-

quently aggregate results generated across different laboratories, time periods, assay variants, and interpretive frameworks. Recent analyses of guideline-conform genotoxicity data have shown that substantial variability and uncertainty remain even within reference datasets assembled from standardized test methods, implying that the apparent performance ceiling of new approach methodologies should be interpreted in the context of intrinsic experimental noise rather than assumed ground truth. More broadly, recent discussions of QSAR predictivity in toxicology have highlighted recurring sources of poor model performance, including deficiencies in data quality, endpoint consistency, mechanistic relevance, uncertainty characterization, and applicability-domain assessment.^{2 2} In other words, improving genotoxicity prediction requires more than a better classifier; it requires a transparent and endpoint-aware data policy.

This need for transparency aligns closely with the FAIR principles, which call for scientific data and associated workflows to be **F**indable, **A**ccessible, **I**nteroperable, and **R**eusable.² In toxicology, FAIR implementation has been framed not only as a data-management ideal but also as a prerequisite for integrating legacy hazard data, new approach methodologies, and computational models into coherent evidence streams.² Recent work has further argued that *in silico* toxicology models themselves should be made FAIR, with explicit documentation of endpoint definition, model provenance, reporting standards, and conditions of reuse to improve reproducibility and regulatory uptake.² This perspective is consistent with the OECD (Q)SAR Assessment Framework, which emphasizes correct model input, applicability domain, prediction reliability, and fitness for purpose, and with recent genotoxicity-specific guidance that stresses the importance of defined endpoints, well-documented predictions, and domain-aware interpretation when applying (Q)SAR approaches in regulatory contexts.^{2 2}

Against this background, data curation and preprocessing are not merely technical preliminaries but scientific decisions that shape the scope and validity of downstream inference. Ames mutagenicity is comparatively well supported by public benchmark resources, which has facilitated broad model development and external comparison.² By contrast, mam-

malian *in vitro* and *in vivo* genotoxicity datasets are often assembled from more heterogeneous evidence streams and therefore place greater demands on explicit endpoint definition, provenance tracking, and rule-based curation.^{???} This distinction motivates an endpoint-specific approach in which structure standardization, salt and metal handling, duplicate resolution, label harmonization, and source annotation are treated as study variables rather than hidden preprocessing defaults.

In the present study, we started from an integrated source workbook that preserved both original test outcomes and identifying metadata, and we used this resource to construct endpoint-specific curated datasets for genotoxicity modeling. For Ames mutagenicity, the final curated dataset combined records originating from the raw industrial corpus with benchmark-derived drug records, while explicitly retaining provenance through a domain variable. In contrast, the mammalian *in vitro* and *in vivo* datasets were curated only from the original source without external benchmark augmentation. We therefore aimed not merely to build another predictive model, but to examine how transparent curation and preprocessing choices affect dataset integrity, comparability across endpoints, and the reliability of downstream *in silico* analyses. Framed in this way, the study addresses a practical question that is highly relevant to both cheminformatics and regulatory toxicology: whether genotoxicity datasets become more reproducible, interpretable, and FAIR-aligned when curation rules are made endpoint-specific, provenance-aware, and explicitly reportable.

Materials and Methods

The raw dataset comprised 397,202 study records and 16,635 identified substances. Study records were mapped to the target genotoxicity endpoints using the property, hazard detail, test guideline, and species fields provided in the source database. Only binary outcomes (positive or negative) were retained, and multiple records for the same substance–endpoint pair were resolved using a predefined priority scheme based on study reliability and study

role, with conservative positive assignment when equivalent conflicting records remained. This procedure reduced the dataset to 331,809 records after binary outcome filtering, 207,011 records after endpoint harmonization, and 179,655 processed records prior to substance-level aggregation. The final substance-level matrix contained 11,852 unique substances, including 11,512 Ames labels, 2,769 in vitro mammalian-cell labels, and 3,479 in vivo somatic-cell labels.

Results

Discussion

Limitations and conclusions

Limitations and conclusions

References

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